

Mahdi Sedaghat

May 2026

Post-Doc at COSIC & Co-Founder at Soundness Labs, Leuven, Belgium

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EDUCATION

KU Leuven <i>Postdoctoral Researcher at COSIC (Part-time)</i> Privacy-Enhancing Techniques in Distributed Systems.	Leuven, Belgium <i>August 2024-Present</i>
KU Leuven <i>Ph.D. Candidate at COSIC</i> Privacy-Enhancing Techniques in Distributed Systems, Supervisor: Prof. Bart Preneel	Leuven, Belgium <i>Jan 2020-July 2024</i>
Sharif University of Technology <i>Master of Secure Telecommunication and Cryptography</i> Attribute-Based Encryption, Supervisors: Prof. MR Aref & Prof. Javad Mohajeri	Tehran, Iran <i>Sept 2015- Sept 2017</i>

EXPERIENCE

Soundness Labs Ltd. <i>Co-Founder, Director at Soundness Labs</i> Post-quantum cryptographic infrastructure for blockchain systems. Designed and implemented ZK proof circuits (SHA-512, Ed25519 verification in zero knowledge), proof aggregation pipelines, and a MAC-based witness bridging protocol. Led applied research on WHIR/FRI-based proving systems and recursive proof composition. Built production Rust and C++ implementations of PQ-ready signature migration tooling.	UK, London (Remote) <i>Jan 2025- present</i>
Computer Science Department, UNSW <i>Visiting Researcher, hosted by Prof. Sushmita Ruj</i>	Sydney, Australia <i>May 2025 - June 2025</i>
Department of Mathematics, Hanyang University <i>Visiting Researcher, hosted by Prof. Jae Hong Seo</i>	Seoul, South Korea <i>May 2025</i>
Foundations of Cryptography, ETH Zürich <i>Visiting Researcher, hosted by Prof. Dennis Hofheinz</i>	Zürich, Switzerland <i>June 2024</i>
Department of Information Engineering, CUHK <i>Visiting Researcher, hosted by Prof. Sherman S. M. Chow</i>	Hong Kong <i>Dec 2023</i>
Mysten Labs. <i>Research Scientist, Internship, Crypto team</i> Contributed to the design and security analysis of the zkLogin protocol—a privacy-preserving blockchain authentication system based on Groth16 zkSNARKs. Served as technical consultant for the Groth16 trusted setup ceremony. Researched Groth-Sahai proof systems and their applications to blockchain primitives.	US (Remote) <i>Apr 2023 - Aug 2023</i>
ZK-Lab, University of Edinburgh <i>Visiting Researcher, hosted by Prof. Markulf Kohlweiss</i>	Edinburgh, UK <i>Feb 2023 - Apr 2023</i>
Computer Science Institute at Charles University <i>Visiting Researcher, hosted by Prof. Pavel Hubáček</i>	Prague, Czech Republic <i>Jan 2019 - Jan 2020</i>

GRANTS

Towards a Quantum Safe Digital Future	<i>Jan 2025 - Present</i>
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Global Seed Fund 2025, joint proposal in collaboration with Prof. Bart Preneel (KUL) and Prof. Sushmita Ruj (UNSW Sydney).

- Developing zk-friendly PQ digital signatures tailored for privacy-preserving technologies (PETs).
- Analyzing and optimizing the performance of network protocols with PQC-based key encapsulation and signature schemes.
- Transitioning authentication mechanisms such as OpenID Connect, OAuth, and TLS to quantum-resistant solutions while maintaining efficiency and compatibility with existing infrastructures.

A Customized Authentication Protocol for zk-Friendly Blockchain Applications June 2025 - Present
Sui SARA Grant with Prof. Bart Preneel (KUL) and Theresa Wakonig (ETH Zurich).

- Create a practical, customizable zk-based authentication method that can influence future standards and be adopted by identity provider consortiums for broader Web3 usability.

OPEN SOURCE PROJECTS

• Longfellow-zk-hiddenpk <i>Post-Quantum protection for all wallet types</i>	C++, Rust 🔗
• PQChain <i>Post-Quantum readiness for EdDSA chains</i>	C++, Rust 🔗
• SP1 proofs on Sui <i>A SP1 Groth16 Proof Verifier for Sui.</i>	Rust 🔗
• Soundness Layer <i>A decentralized verification layer for ZK proofs.</i>	Rust 🔗
• Unlinkable Policy-Compliant Signatures <i>Prototyping the PCS and several implementations for ul-PCS schemes.</i>	Python, Docker 🔗
• Groth-Sahai Proofs <i>An efficient implementation for the seminal work of Jens Groth and Amit Sahai proof system.</i>	Python 🔗
• Nirvana Payment <i>A distributed implementation of an anonymous and reusable payment guarantee system.</i>	Python 🔗
• Cross-Domain Attribute-Based Access Control Encryption (CD-ABACE) <i>Proof of concept for the cross-domain access control encryption scheme.</i>	Python 🔗

TECHNICAL SKILLS

- **Rust:** Production cryptographic implementations—built ZK proof verifiers (SP1/Groth16 on Sui), a decentralized proof verification layer, MAC-based witness bridging protocols, and a threshold FROST MPC signing demo with Axum backend. Daily driver for protocol-level and performance-critical code.
- **C++:** ZK circuit development—implemented full SHA-512 and Ed25519 arithmetic circuits for the PQChain proving system, including field arithmetic alignment across BN254 and Ed25519 scalar/base fields.
- **Python / SageMath:** Rapid prototyping of cryptographic constructions—Groth-Sahai proofs, unlinkable policy-compliant signature schemes, anonymous credential systems, and pairing-based primitives. Used extensively for security experiments and paper artifact implementation.
- **Move:** Smart contract development on Sui—WHIR proof verifier prototype deployed to testnet, SP1 Groth16 on-chain verification contracts.
- **Proof Systems:** Hands-on experience with SNARK/STARK internals: Groth16 trusted setup ceremonies, FRI/STIR/WHIR proximity-testing-based proof systems, Reed-Solomon and algebraic geometry codes, recursive proof composition, and proof aggregation architectures.
- **Infrastructure:** React/Node full-stack development, self-hosted Linux server administration (nginx, systemd, certbot), Docker, CI/CD pipelines.

STUDENTS & TEACHING EXPERIENCES

- **Hossein Moghaddas** (PhD student, co-Supervisor with Prof. Bart Preneel)
- **Hossein Hafezi** (Visitor -> Cambridge University)
- **Kiran Deep Ghosh** (ISI Kolkata -> RUB), master thesis supervisor.
- **Theresa Wakonig** (ETH Zurich), master thesis co-supervisor.
- **Internship mentoring:** Anonymous Credentials, Student: Peter Schwarz, COSIC, KU Leuven (2023).
- **Lecturer** in Privacy course on Anonymous Credential systems, COSIC, KU Leuven (2022-2023).
- **Mentoring** in CyberSecurity Basics course, COSIC, KU Leuven (2022-2023 & 2023-2024).
- **Internship mentoring:** Decentralized e-Voting systems, Student: Sermin Kocaman, COSIC, KU Leuven (2022).
- **Master's Thesis Supervision:** Privacy assessment of current business practices using blockchains in banking and financial sector, Jowhar Ding, COSIC, KU Leuven (2020-2021).

PROFESSIONAL SERVICE

I am serving as a PC member at:

- **ACM CCS 2026**, as ZK proof systems reviewer.
- **Information Security Conference** (ISC 2025, 2026).
- **PrivCrypt-2026** workshop, co-located with EuroCrypt-2026.
- **PrivCrypt-2025** workshop, co-located with ACNS-2025.

I have served on the **TIFS-2025, PETS-2025, AC-2024, CANS-2024, CRYPTO-2024, PKC-2024, IEEE TDSC-2024, LatinCrypt-2023, ACM CCS-2023, IEEE TDSC-2023, IEEE TIFS-2022, EC-2022, AC-2020, TCC-2019** and **ISCISC-2018** as reviewer.

AWARDS AND ACHIEVEMENTS

- The best proposal for the Virtual design challenge for authentication and protecting Full Motion Video system, University of British Columbia, Canada, 2019. 
- Ranked 46th in M.Sc. national university entrance exam in Communications branch among about 20,000 participants, 2015.
- Ranked 36th in Iranian National Olympiad in Electrical Engineering among all bachelor students of Electrical Engineering, 2014.

EXTRA

- Soundness Blogpost, 
- Blogpost, SP1 Verifier on Sui 
- Blogpost, Eurocrypt 2024: Twinkle (A Fully Adaptive Threshold Signature from DDH) 
- Blogpost, Groth-Sahai Proofs: Zero to Hero. 
- Technical consultant in the Groth'16 Ceremonial Setup for zkLogin project at Mysten Labs. 

TALKS

- 4th Edition of the TUM BlockchainCybersecurity Salon, Munich, Germany, 2-3 July 2026.
- Post-Quantum Blockchain Security, zkSummit 14, Rome, Italy, 07 May 2026.
- Post-Quantum Readiness in EdDSA Chains, Financial Crypto 2026, St. Kitts, 04 March 2026.
- Zero-Knowledge proofs: Applications to Blockchain, Cybersecurity Industry Day 2024, Mechelen, Belgium.
- zkLogin, Foundations and Applications of Zero-Knowledge Proofs, Edinburgh UK, 04 Sept 2024.

- Unlinkable Policy-Compliant Signatures for Compliant and Decentralized Anonymous Payments, Privacy-Enhancing Technologies Symposium (PETS) 2024 in Bristol, UK, 18 July 2024.
- Threshold Structure-Preserving Signatures: Done and Ongoing Projects, Foundations of Cryptography, ETH Zürich, Switzerland, 04 June 2024.
- Subset-optimized BLS Multi-signature with Key Aggregation, Financial Crypto 2024, Curacao, 5 March 2024.
- Unlinkable Policy-Compliant Signatures for Compliant and Decentralized Anonymous Payments, CUHK, Hong Kong, 12 Dec 2023. [link](#)
- Threshold Structure-Preserving Signatures, Asiacrypt, Guangzhou, China, 6 Dec 2023. [link](#)
- Trusted Setups for zkSNARKs, Mysten Labs Paris offsite, 4 August 2023.
- Unlinkable Policy-Compliant Signatures, Blockchain Technology Lab (BTL), Edinburgh, 20 March 2023.
- Cross-Domain Attribute-Based Access Control Encryption, CANS'21, Online, 13 December 2021.

Publications

Foteini Baldimtsi, Konstantinos Chalkias, Arnab Roy, and Mahdi Sedaghat. Post-quantum readiness in EdDSA chains. *Financial Crypto 2026*, 2025.

Jan Bobolz, Emad Heydari Beni, Anja Lehmann, Omid Mirzamohammadi, Cavit Özbay, and Mahdi Sedaghat. Multi-verifier keyed-verification anonymous credentials. *Cryptology ePrint Archive*, Paper 2025/2156, 2025.

Karim Bagheri, Ehsan Ebrahimi, Omid Mirzamohammadi, and Mahdi Sedaghat. Traceable verifiable secret sharing and applications. *Cryptology ePrint Archive*, Paper 2025/318, 2025. <https://eprint.iacr.org/2025/318.pdf>.

Omid Mirzamohammadi, Jan Bobolz, Mahdi Sedaghat, Emad Heydari Beni, Aysajan Abidin, Dave Singelee, and Bart Preneel. Keyed-verification anonymous credentials with highly efficient partial disclosure. *CiC 2025*, 2025. <https://eprint.iacr.org/2025/041.pdf>.

Mahdi Sedaghat and Bart Preneel. Privacy-Enhancing Techniques in Distributed Systems. *PhD Thesis*, 2024. <https://cosicdatabase.esat.kuleuven.be/backend/publications/files/these/514>.

Foteini Baldimtsi, Konstantinos Kryptos Chalkias, Yan Ji, Jonas Lindstrøm, Deepak Maram, Ben Riva, Arnab Roy, Mahdi Sedaghat, and Joy Wang. zkLogin: Privacy-Preserving Blockchain Authentication with Existing Credentials. *ACM CCS'24 and presented at SBC'24, NY, and RWC'25, Sofia*, 2024. <https://arxiv.org/pdf/2401.11735>.

Christian Badertscher, Mahdi Sedaghat, and Hendrik Waldner. Fine-Grained Accountable Privacy via Unlinkable Policy-Compliant Signatures. *Cryptology ePrint Archive*, Paper 2023/1070 (PETS'24 and presented at CTB workshop at EC'24), 2023. <https://eprint.iacr.org/2023/1070>.

Aikaterini Mitrokotsa, Sayantan Mukherjee, Mahdi Sedaghat, Daniel Slamanig, and Jenit Tomy. Threshold Structure-Preserving Signatures: Strong and Adaptive Security Under Standard Assumptions. In Qiang Tang and Vanessa Teague, editors, *Public-Key Cryptography – PKC 2024*, pages 163–195, Cham, 2024. Springer Nature Switzerland.

Foteini Baldimtsi, Konstantinos Kryptos Chalkias, Francois Garillot, Jonas Lindstrom, Ben Riva, Arnab Roy, Mahdi Sedaghat, Alberto Sonnino, Pun Waiwitlikhit, and Joy Wang. Subset-optimized BLS Multi-Signature with Key Aggregation. *Cryptology ePrint Archive*, Paper 2023/498 (Financial Crypto 2024), 2024. <https://eprint.iacr.org/2023/498>.

Elizabeth Crites, Markulf Kohlweiss, Bart Preneel, Mahdi Sedaghat, and Daniel Slamanig. Threshold Structure-Preserving Signatures. In *International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT'23)*, pages 348–382. Springer, 2023. <https://eprint.iacr.org/2022/839>.

Karim Baghery, Axel Mertens, and Mahdi Sedaghat. Benchmarking the Setup of Updatable Zk-SNARKs. In *Progress in Cryptology – LATINCRYPT 2023*, pages 375–396, Cham, 2023. Springer Nature Switzerland. <https://eprint.iacr.org/2023/1161>.

Akash Madhusudan, Mahdi Sedaghat, Samarth Tiwari, Kelong Cong, and Bart Preneel. Reusable, Instant and Private Payment Guarantees for Cryptocurrencies. In *Information Security and Privacy - 28th Australasian Conference, ACISP 2023, Brisbane, QLD, Australia, July 5-7, 2023, Proceedings*, volume 13915 of *Lecture Notes in Computer Science*, pages 580–605. Springer, 2023. <https://eprint.iacr.org/2023/583>.

Seyed Farhad Aghili, Mahdi Sedaghat, Dave Singelee, and Maanak Gupta. MLS-ABAC: Efficient Multi-Level Security Attribute-Based Access Control scheme. *Future Generation Computer Systems*, 2022. <https://www.sciencedirect.com/science/article/pii/S0167739X22000115>.

Karim Baghery and Mahdi Sedaghat. Tiramisu: Black-Box Simulation Extractable NIZKs in the Updatable CRS Model. In *Cryptology and Network Security (CANS)*, pages 531–551, Cham, 2021. Springer International Publishing. <https://eprint.iacr.org/2020/474>.

Mahdi Sedaghat and Bart Preneel. Cross-Domain Attribute-Based Access Control Encryption. In *Cryptology and Network Security (CANS)*, pages 3–23. Springer International Publishing, 2021. <https://eprint.iacr.org/2021/074>.

Reyhaneh Rabaninejad, Seyyed Mahdi Sedaghat, Mohamoud Ahmadian Attari, and Mohammad Reza Aref. An id-based privacy-preserving integrity verification of shared data over untrusted cloud. In *2020 25th International Computer Conference, Computer Society of Iran (CSICC)*, pages 1–6, 2020.